

POSTDOCTORAL RESEARCHER (COMPUTATIONAL GENOMICS)

Pacific Biosciences Research Center Pipes Laboratory

- I. **SUMMARY OF POSITION:** Regular, Full-Time, RCUH Non-Civil Service position with the Pacific Biosciences Research Center (PBRC), Pipes Laboratory, located on the University of Hawai'i at Mānoa (UHM) in Honolulu, Hawai'i. Continuation of employment is dependent upon satisfactory work performance, program/operational needs, availability of funds, and compliance with applicable Federal/State laws.
- SUMMARY OF DUTIES:** Develops and implements novel statistical phylogenetic algorithms and high-performance computing software for analyzing ultra-large evolutionary trees. Focuses on advancing likelihood and Bayesian inference frameworks to solve complex problems in environmental DNA and wastewater surveillance, such as phylogenetic placement and ancestral state reconstruction. Engineers scalable, memory-efficient solutions in C/C++ to handle massive datasets. Prepares manuscripts for peer-reviewed journals, presents findings at conferences, and mentors students in computational best practices.
- II. **SCOPE OF POSITION:**
- A. **Reports to:**
- | | |
|------------------------|--------------|
| Principal Investigator | Lenore Pipes |
|------------------------|--------------|
- B. **Supervises:** None.
- C. **Budgetary and/or Fiscal Responsibilities (applicable for only positions responsible for budgeting, payroll or procurement):** None.
- D. **RCUH Human Resources or Financial Portal Access and Action Authorization Levels:** None.
- E. **Level of Interaction:** The individual shall work closely together with the Principal Investigator (PI) on decision making regarding research direction but independently on planning and carrying out the experiments. They will develop projects with progressively increasing levels of independence. They will attend and participate in regular lab/project meetings, and interact with supervisors, co-workers, and other lab personnel respectfully and professionally. Work environment will be a combination of office setting, wet lab and field travels.

III. **MAJOR DUTIES & RESPONSIBILITIES (See column instruction below):**

IIIA = % time/effort

IIIB = Place X if job duty is an Essential Job Function

IIIC = Job duty number (**BOLD if Primary Duty**)

IIID = Narrative description of Job Duty (**BOLD if Primary Duty**).

IIIA	IIIB	IIIC	IIID
% of Time	Essential Job Function	No.	Description of Major Duty. BOLD if Primary Duty.
45%	X	1	Development of Statistical Phylogenetic Algorithms: Designs and implements advanced algorithms for phylogenetic inference (ML, Bayesian), phylogenetic placement, and ancestral state reconstruction.
25%	X	2	High-Performance Computing and Software Implementation: Develops and optimizes memory-efficient software programs (primarily C/C++) for large-scale data.
15%	X	3	Writes and submits research articles for peer-reviewed journals and presenting findings at conferences, seminars, and lab meetings.
10%		4	Collaborates with other researchers, participating in lab meetings, and mentoring graduate/undergraduate students and other research staff on computational techniques and data analysis.
5%		5	Any additional responsibilities related to lab management, management of the server, project development, or administrative tasks.
		6	Performs other duties as assigned.

IV. **PRIMARY QUALIFICATIONS:**

	Qualification	Description
A	EDUCATION	PhD from an accredited college or university in Statistics, Computer Science, Computational Biology, or a related quantitative field.
B	EXPERIENCE	One to three (1-3) years of experience in developing algorithms or statistical models for evolutionary analysis. Experience writing production-level code is highly preferred.
C	KNOWLEDGE	Expert-level proficiency in C/C++ (preferred). Strong theoretical understanding of statistical phylogenetics, including substitution models, likelihood theory, Bayesian MCMC, and coalescent theory. Knowledge of graph algorithms and computational complexity regarding tree topology search. Familiarity with HPC environments.
D	ABILITIES & SKILLS	Proven ability to independently design and implement mathematical models for evolutionary processes. Strong experience in debugging complex numerical

		optimization codes. Must be able to work independently & manage multiple tasks effectively and troubleshoot complex software issues. Post-Offer/Employment Conditions: Must be able to complete the UH Title IX training within twelve (12) months from date of hire. Must be able to complete the UH Information Security Awareness Training (ISAT) within two (2) weeks from date of hire, and re-certify every twelve (12) months.
E	PHYSICAL/MEDICAL DEMANDS	Ability to work at a computer for extended periods, performing tasks that require sustained focus, attention to detail, and precision.
F	POLICY/REGULATORY REQUIREMENT	As a condition of employment, employee will be subject to all applicable RCUH policies, procedures, and trainings and, as applicable, subject to University of Hawai'i's and/or business entity's policies, procedures, and trainings. Violation of RCUH's, UH's, or business entity's policies and/or procedures or applicable State or Federal laws and/or regulations may lead to disciplinary action (including, but not limited to possible termination of employment, personal fines, civil and/or criminal penalties, etc.).

V. SECONDARY QUALIFICATIONS:

Demonstrated ability to write scientific papers independently. Strong verbal communication skills for presenting research at conferences, seminars, and lab meetings.